

Tesla EV Growth Strategy · Washington EV Hub

- <https://home-page-ev.onrender.com/>

Why the EV Forecasting System is Useful

The EV transition depends not only on vehicle availability but also on building charging infrastructure that reduces range anxiety and enables widespread adoption. Washington State's climate goals—reducing greenhouse gas emissions by 45% by 2030, 70% by 2040, and 95% by 2050—require a significant increase in EV adoption. This system helps policymakers and planners understand how charging infrastructure, population density, and EV demand interact.

1. Visualizing Population Density vs. Charging Availability

The statewide map provides a clear comparison between population density and existing supercharger locations. This visualization highlights:

- Dense counties where charging availability is insufficient
- Areas where high population + low infrastructure = high risk of charging anxiety
- Disproportionate infrastructure distribution created by corridor-first planning

This allows planners to immediately identify counties where infrastructure investment will have the greatest impact on EV adoption.

2. Budget Allocation and County Prioritization

Because EV adoption is strongly influenced by charging accessibility, the system supports strategic budget allocation for charger expansion. Agencies or companies like Tesla can:

- Allocate infrastructure budgets based on population density
- Prioritize counties where chargers yield the largest EV adoption increases
- Compare ROI of charger deployment between counties

The county-level tool provides a granular understanding of geographic demand, enabling statements like:

“County A requires a larger share of the statewide charging budget because population density indicates high demand, while County B requires fewer chargers but will see major benefits from even small expansions.”

3. Forecast Curves Predict Charging Needs to Meet Policy Goals

The system uses pre-generated cubic and logistic forecasting formulas derived from Monte Carlo simulation and time-series scaling. These formulas allow users to model:

- How EV registrations change with each additional charger
- How adoption rates shift across counties as infrastructure improves
- How many chargers are needed to meet 2030–2050 climate mandates
- Whether certain counties are on track or at risk of falling behind

Examples of insights:

- “Adding 20 superchargers in King County produces a measurable increase in EV registrations.”
- “Washington must build over 100 additional superchargers statewide by 2030 to meet EV adoption targets.”

This transforms the tool into a scenario-planning and policy-evaluation engine.

4. A Practical Tool for Planners, Utilities, and EV Strategists

By combining Monte Carlo results, curve-fitting formulas, GIS mapping, and interactive forecasting, the system provides a unified, accessible decision-support platform. It enables:

- Transparent communication of infrastructure needs
- Data-driven planning for charger expansion
- Identification of underserved populations
- Practical forecasting for policymakers and industry stakeholders

Summary

The EV Forecasting System is useful because it:

- Identifies infrastructure gaps using population and charger data
- Guides budget planning and county prioritization
- Predicts how charging expansion drives EV adoption
- Helps meet state climate-policy requirements
- Provides a centralized forecasting platform inspired by Tesla’s infrastructure strategy

It turns complex modeling into actionable guidance for real-world EV infrastructure planning.

EV Forecasting System – Deployment Documentation

1. API Documentation

This documentation describes the deployed Flask applications - Washington EV Forecasting System, which includes a Home Hub UI, a Statewide Forecast Tool, and a County-Level EV Forecast Tool. All formulas used for forecasting were pre-generated during the Monte Carlo scaling phase and exported into Excel. The deployed Flask applications simply evaluate these formulas based on user inputs.

1.1 Base Routes

GET / — Home Hub where users choose between the statewide or county-level tool.

1.2 Statewide Forecast Tool

Routes:

GET /statewide

POST /statewide

Users input the number of supercharger points. The app loads the statewide cubic formula from Excel and evaluates EV registration and adoption rate predictions.

static/data/County-level formula_cleaned.xlsx

1.3 County-Level Forecast Tool

Routes:

GET /county

POST /county

Users enter a county name and a number of supercharger points. The tool retrieves the corresponding pre-generated cubic and logistic formulas from Excel and computes EV registration and adoption rate predictions.

static/data/County-level formula_cleaned.xlsx

1.4 Static Assets Endpoints

/chart/<filename> ##which is county name

/map/<filename> ##which is county name

Serves pre-rendered PNG charts and GIS maps for statewide and county-level visualization.

1.5 Forecast Evaluation Engine

Formulas used in the app are not generated dynamically. They were pre-calculated during model expansion phase using cubic regression (EVs vs Supercharger and adoption). The Flask engine safely evaluates these formulas for any X value entered by the user.

static/data/County-level formula_cleaned.xlsx

python

 Copy code

```
def evaluate_formula(formula_str, x_value):  
    """  
    Evaluates a pre-computed polynomial/logistic forecasting formula.  
  
    These formulas were generated previously during the model scale-up phase  
    using cubic regression (EVs vs Superchargers) and logistic fits  
    (Adoption Rate vs Superchargers), and then exported into Excel files.  
  
    Parameters  
    -----  
    formula_str : str  
        Cleaned mathematical expression from the Excel file.  
        Example: "y = 0.0021*x**3 - 4.7*x**2 + 4500*x - 140000"  
  
    x_value : float  
        Number of supercharger points input by user.  
  
    Returns  
    -----  
    float  
        The predicted EV registration or adoption value.  
    """
```

2. Installation & Build Instructions

This project contains three separate Flask applications:

1. Home Hub UI
2. Statewide Forecast Tool
3. County-Level Forecast Tool

```
/home_hub_app/  
  app.py  
  templates/index.html  
  static/
```

```
statewide_app/  
  app.py  
  templates/index.html  
  static/data  
  static/maps  
  static/charts
```

```
county_app/  
  app.py  
  templates/index.html  
  static/data  
  static/maps  
  static/charts
```

2.1 Clone the Repository

```
git clone https://github.com/judyc4986/ev_home_hub  
cd /Desktop/ev_home_hub
```

```
git clone https://github.com/judyc4986/ev_render_app  
cd /Desktop/ev_render_app
```

```
git clone https://github.com/judyc4986/ev_forecast_app  
cd /Desktop/ ev_forecast_app
```

2.2 Set Up Virtual Environment

```
python3 -m venv venv  
source venv/bin/activate
```

2.3 Install Requirements

```
pip install -r requirements.txt
```

2.4 Running Applications Locally

Each UI runs independently. Example:

```
cd ev_home_hub  
python app.py
```

```
cd ev_render_app  
python app.py
```

```
cd ev_forecast_app  
python app.py
```

Access at **<http://127.0.0.1:5000/>**

2.5 Deployment on Render

In Render add webservice setting, link to git repository and select command:

https://github.com/judyc4986/ev_home_hub
https://github.com/judyc4986/ev_render_app
https://github.com/judyc4986/ev_forecast_app

Build Command:

```
pip install -r requirements.txt
```

Start Command:

```
gunicorn app:app
```

Build & Deploy

Repository

The repository used for your Web Service.

https://github.com/judyc4986/ev_home_hub

 Edit

Branch

The Git branch to build and deploy.

main



 Edit

Git Credentials

User providing the credentials to pull the repository.

judyc4986@gmail.com (you)

Use My Credentials

Build Command

Render runs this command to build your app before each deploy.

```
$ pip install -r requirements.txt
```

[Edit](#)

Pre-Deploy

Command Optional

Render runs this command before the start command. Useful for database migrations and static asset uploads.

```
$
```



[Edit](#)

Start Command

Render runs this command to start your app with each deploy.

```
$ gunicorn app:app
```

[Edit](#)

3. Demonstration

A demonstration of screenshots to show how to use the statewide and county-level tools. The notebook documents user workflow.

1. Screenshot of Home Hub UI (choose statewide or county-level)
<https://home-page-ev.onrender.com/>

WA CLIMATE · TESLA EV MISSION

SCENARIO MODELING FOR EV ADOPTION & CLEAN-ENERGY TRANSITION

Washington EV Adoption Simulator & Tesla EV Growth Strategy Alignment

This hub blends Washington State's climate policy goals with a Tesla-inspired approach to EV adoption — modeling how charging availability, county adoption differences, and infrastructure rollout speed help accelerate the transition to sustainable energy.

Built by **Judy Cheng** — data-driven simulations for **state planners**, **utilities**, and **EV strategists**.

Explore the Tools

Choose a lens: county-level sensitivity or statewide EV infrastructure modeling.

TOOL 01 · COUNTY-LEVEL

EV Registration vs Supercharger

Explore how each county's EV adoption responds to additional superchargers. Identify where chargers enable the largest incremental growth in EV registrations.

Open County Tool

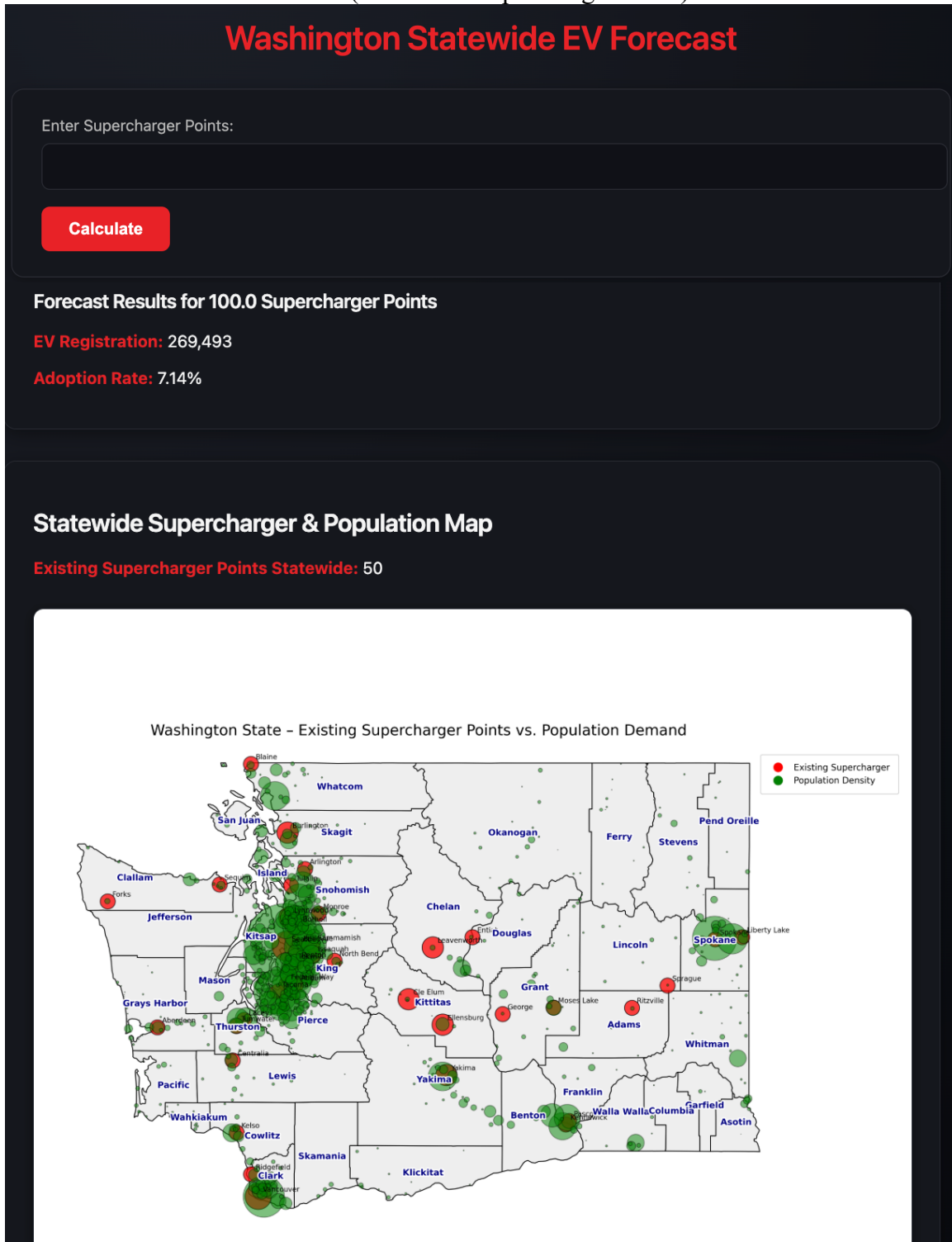
TOOL 02 · STATEWIDE

EV Registration vs Supercharger

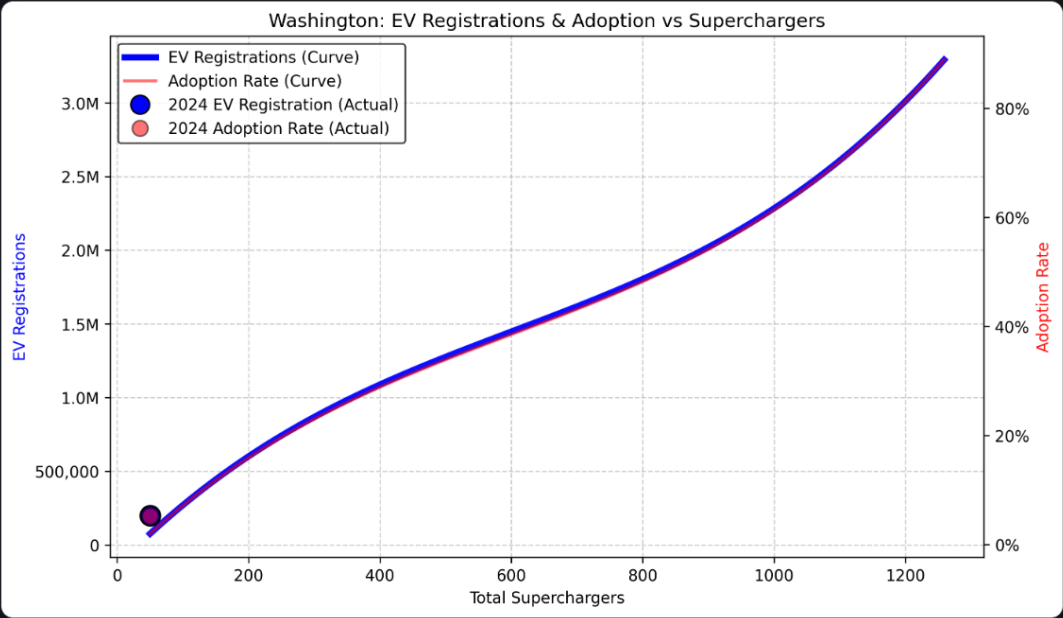
Model statewide EV adoption pathways under different charging build-out assumptions, matching Washington's climate goals and Tesla-style infrastructure strategy.

Open Statewide Tool

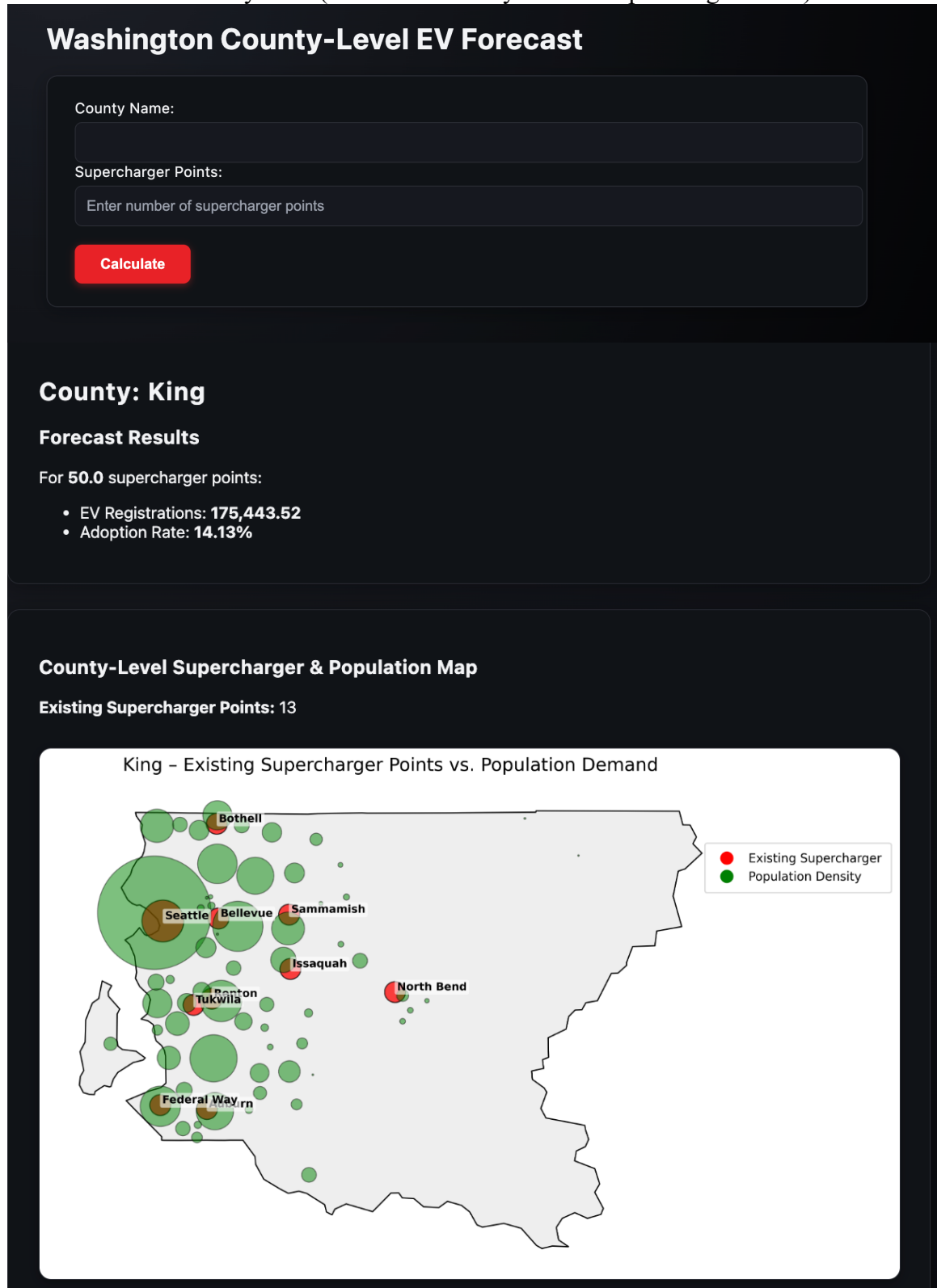
2. Screenshot of Statewide Tool (user enters supercharger count)



Statewide Forecast Curve



3. Screenshot of County Tool (user enters county name + supercharger count)



County-Level EV Forecast Curve

